



DEALER BEST PRACTICE SERIES

Parts Management

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1.0 Introduction

Parts Management is a critical logistical function that supports the Scheduled and Unscheduled Maintenance and Repair activities. A properly managed parts inventory and procurement system is fundamental to achieving fleet availability targets in the most cost-effective manner. Accurate and complete demand history and a well-supported forecast of future needs are key elements that enable the Parts Department to have the correct parts, in the correct quantities and at the correct time. To accomplish this, the parts, planning and service groups need to coordinate their activities and share information related to all repairs and repair plans.

This Best Practice outlines the implementation of the Parts Management function on site to effectively support planned Maintenance and Repair as well as unscheduled repair events.

2.0 Best Practice Description

Outlined in the following sections are the ingredients that support an efficient parts warehouse operations is being maintained at the mine site.

Description Outline for Parts Management

- 2.1 Parts Management Strategy
- 2.2 On-Site Warehouse
- 2.3 Parts Inventory Management
- 2.4 Component Inventory Management
- 2.5 Backlog Parts Management
- 2.6 Tool Management
- 2.8 Human Resources
- 2.9 Performance Measurement

2.1 Parts Management Strategy

The basis of any successful Parts Management (PM) process is a documented and complete strategy. The strategy will be the overall reference document that defines what the organization intends to accomplish and a plan of how the strategy will be executed. We begin to form the plan by clearly defining the goals and objectives of the Parts Management process. These goals and objectives need to link with the Maintenance Strategy, Planning department, site characteristics and in country logistics. Many other factors may influence the Parts Management strategy depending on the repair habits and in country business policies.

- Provide 24 / 7 support to the daily on-site M&R operation
- Manage efficient & effectively the on site inventory
- Maintain effective Parts / Components Supply channels
- Work closely with the Planning & Scheduling area to provide the adequate Parts & Components support to the Scheduled Activities
- Support with Parts & Components the unscheduled events. Meet planning expectations / goals

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- Manage efficient & effectively the additional related resources: Tools, Instruments, Consumables, etc.
- Maintain and submit updated information of Parts Management

2.2 Warehouse Operations and Support

The location, size and hours of operation of the warehouse are important. The location of the warehouse should be within a short walking distance of the repair shop. If at all possible, parts should be delivered to the bays or to a drop area specified in the shop by a parts runner. This will insure the technicians spend their time servicing machines in the bays and not searching for parts.



*PM Parts delivered at the PM Bay in advance of the PM service
Same location ... & on time*

Storage of the parts should be efficient and orderly. The received parts should be put away in their location daily to insure inventory accuracy. Location of frequently used parts such as maintenance items should be near the front of the parts warehouse for easy retrieval.

Warehouse layout and storage are important to Warehouse Support. Adequate warehouse space is essential. As density of parts increase, operational efficiency falls. Part numbers per 1000 square feet is a good measure for density.

Calculate the parts numbers per 1000 square feet and follow the guidelines below to confirm if warehouse rearranging or addition is necessary.

- ❑ <1000: Probably too much warehouse space
- ❑ =1000: Used for new warehouse design
- ❑ 1001-1500: Indicates a warehouse comfortably full but not stressed
- ❑ 1501-2000: Indicates a need to provide new warehouse space.
- ❑ 2001-2500: Warehouse is approaching gridlock, insufficient space to operate efficiently.
- ❑ >2500: The warehouse should be impossible to move around in, and aisles are full.

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Methods of storage and location of inventory by class are important considerations. The following items should be considered.

- What type of storage equipment is being used? For dense storage and to maximize warehouse storage efficiency, we recommend the use of storage cabinets (Vidmar). This is the number one storage concept that is recommended. Sixty to seventy-five percent of Cat parts that are inventoried can be stored in a cabinet. The cabinet concept allows storage that is easily accessible at the ground level without lifts. The location should be near the front of the parts department where it can be easily picked.
- The second storage concepts recommended are bins (15% to 20% of Cat parts are stored in bins), which have a 24" depth, 36" depth or 48" depth.
- Racking is then the third major storage concept.
 - o For cabinets and bins especially, only one part should be stored in a given location. Although dealers are storing more than one part on a given section of racking, we encourage them to assign multiple locations to a rack section depending on the number of items being stored.
 - o The faster moving items (items with high demand based on calls) should be stored near the main aisles or sections of cabinets & bins. In other words, high demand items should be stored close to the work area.
- Is enough floor space available to store major components within the warehouse? Store these components inside, if possible because of the outside elements.
- What type of warehouse location system is used or is planning to use? The location system should be easy to understand and can be learned in 20 minutes. Should include aisle, section, elevation & opening.
- Specific areas of the warehouse should be designated for PM kits, Backlog parts, Component installation kits that are used for repetitive repairs/maintenance for easy access.
- The method used in parts storage should follow Cat Contamination Control guidelines.

The hours of operation of the warehouse should match that of the repair shop, 24/7.

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2.3 Parts Inventory Management

Inventory Control Management is simply providing the right parts in the right quantity to the right place, at the right time at the lowest cost. Parts become stocked because of either anticipated demand or historical demand. Once a part has been added to stock, it may be controlled in a variety of ways, but will typically have a minimum and maximum setting. Minimums define “when” to order while maximums help define “how much” to order.

Lead-time is a critical factor with minimum vs maximum. Lead-time is the total number of elapsed days from when a part reaches order point to when the resulting stock order is received, put away (physically and in the system) and made available for sale. If overstated, it may create needless surplus of stocked items, eventually having a negative impact on turnover. If understated, it may cause stock backorders that will impact customer service. Lead-time is affected by country location, import restrictions, and local transportation.

The key to having the correct parts available is working closely with the planning and repair management functions of the site operations. Knowledge of the fleet make-up and repetitive maintenance and/or repairs play an important role in maintaining inventory. Standard job usage helps to identify parts needed for service work. Pre-ordering the parts allows sufficient levels of inventory to be maintained. Some parts take longer to acquire than others. Maintaining communication with the Backlog management group is equally important so jobs can be released for scheduling when the parts become available.

Another important ingredient to the ordering process is having one central location identified in the warehouse where the parts orders from the repair shop are received. This may be a computer terminal or in-basket. This limits the overlooked parts orders. Receiving and

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inventorying of the parts daily is important. Having the parts on the shelf and in the inventory system will minimize inventory carrying cost by eliminating duplication of unnecessary parts.

Consumables and high rotation items need to be readily available. This can be merely having bins located in the repair shop with seals, nuts and bolts. Their levels need to be monitored to insure a good supply is kept available. An accounting method will need to be developed to insure the items get allocated fairly to the work orders. Field service strategy also requires similar strategy in order to reduce the waiting time for field technicians.



Contamination Control plastic plugs stored in a contamination free cabinet (CGM Best Practice published)

Proper fluids handling will need to be determined. Storage and dispensing of the fluids will need to be safe and according to local safety regulations. Contamination Control plays an important role in the storing and dispensing of the fluids. Likewise, waste management of the used fluids is important so not to contaminate the environment.

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The use of PM parts kits will also increase the efficiency of parts management. PM kits and rebuild kits should be used for the most frequent “known” routine and repairs. The kits are built ahead of time, off site, based on the planning and scheduling activities. All the parts are collected in a “box” that are associated with the repair. The kit is delivered in its entirety to the technician. Kits should not be built for less frequent repairs because part numbers may change and additional labor would be required to update the kits. Parts and kits should also be made available to the field technicians in the same manner as they are for shop technicians.



PM kits packing example
Kits in the process of being prepared



PM Kits stored on site

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2.4 Component Inventory Management

An adequate Component Management plan plays an important role in the overall success of the operation, keeping costs and downtime under control. This is extremely important when equipment enters the component exchange interval that adequate component availability exists. Component Inventory Management involves a number of activities. These activities complement one another and typically include:

- Component tracking.
- Component storage and transportation controls.
- Fluent communications between the site and the rebuild center.
- Well-defined and updated planned component replacement strategy (PCR).
- Measurement and tracking of carrying costs related to the Component Management strategy.

Keeping an up-to-date inventory with the proper amount of components that matches the fleet requirements and schedules is important. Inventory includes protective and normal stock for supporting the Planned Component Repairs (PCR's). Strategy for handling this inventory is important also. Are the components going to be dedicated to the particular site or as part of a pool of components available for multiple sites? Coordination with the rebuild center and planning/scheduling is critical based on the Long Term and Medium Term repair plans. Rotation of stock is important in order to make the latest engineering design changes available.. Cores need to be properly identified when taken out of equipment so tracking of hours to track hours of operation and failure mode is accurate.

By effectively performing these on-site activities, it is possible to manage components within the defined goals. Clearly there are other support functions that must exist that go beyond on-site operations, e.g. Component Rebuild Center and spare parts / component support. Thus the functionality of site support software should focus on the on-site activities as well as the interaction between the site and those outside support entities.

Core Management is consistently an area of significant financial opportunity whether speaking of component cores or warranty cores. Cores need to be returned and properly tagged so the parts department can disperse them properly for rebuilding or warranty credit. Some elements of core handling are identified below and need to have processes defined:

- Who is responsible for inspecting worn cores? One person?
- Is warranty core inspection accuracy at least 99%? Is full core credit issued when the parts department says it should?
- Do regular processes exist to physically locate all remaining worn undistributed cores and update inventory and general ledger if they can't be found?
- Are core processing areas neat, organized and efficient?
- Are cores cased and staged in racks?

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2.5 Backlog Management

The backlog can take many forms. It can be a simple list of expected jobs or it can be a series of open work orders in a computerized database. When the computer system has a quick and easy method of opening work orders, it is the preferred method, parts can be ordered ahead of time to eliminate delays.

A specific area in the warehouse needs to be identified for storing parts for backlogs until the equipment is scheduled or the entire list of parts is available. The inventory needs to be protected from technicians helping themselves. This also holds true for backordered parts identified for an open work order.

Once the complete inventory of parts is available, the person(s) responsible for planning and repair management needs to be contacted so the work can be scheduled. Following the Backlog process map logic where colored logs are used, the paper copy would move from the Blue Phase to the Green Phase.

2.6 Tool Management

Providing specialized tools and test instruments is crucial to maintaining a fleet. This inventory of tools is expensive and requires special handling. The tools are usually kept in the parts department and inventoried. A check-out and check-in procedure needs to be established so the tools don't disappear or end up locked in a personal toolbox. Repairs to the tooling need to be identified in a timely fashion. Completing a tag that identifies needed repairs upon check-in will serve the purpose. Regular testing of the tooling and gauges is required on a scheduled basis. Proper tagging or labeling of the tooling should be required that indicates the date last tested. One person should be made responsible for tool management.

2.7 Human Resources

Maintaining the Parts Operations positions with qualified staff is important to fulfilling the strategy identified and assuring the success of the process. Listed are the recommended positions and associated activity:

- Parts Analyst is the on site leader of the Parts/Components operation
- Warehouse Operation staff must cover 100% of shop – field operations
- Tool Room Attendant keeps specialized tools and test instruments available and in good condition
- Parts Runner supports efficiency of the service operations by taking parts to the shop and returning used and unused parts/cores to the parts department.
- Local Purchasing is a must in acquiring local needed resources and maintaining an efficient local supplier network

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2.8 Performance Measurement

The use of metrics should be considered when managing a parts operation. They are used to determine how well maintenance and repair activities are supported by the parts inventory and evaluate the relationship between the Parts Operations, Planning and Scheduling, and Maintenance Departments in their efforts to avoid unnecessary parts related downtime. Please reference section 2.5 in the Metrics (KPI's) to Assess Process Performance written by A. Flores and J. McCaherty. Contact A. Flores or J. McCaherty to receive a copy.

Warehouse Service Fill Level (instantaneous)

$$\text{Service Fill Level} = \frac{\text{Parts Orders Closed at 1st Request}}{\text{Total Parts Orders}} \times 100 \quad (\%)$$

(instantaneous)

Instantaneous Service Fill level is a parts management efficiency indicator that quantifies the percentage of individual parts requests entered against the on-site parts warehouse for repairs (including Backlog parts requests) and filled/closed at the first call. A reflection of the level of satisfaction of on-site parts warehouse performance. The Benchmark for Instantaneous Service Fill Level is 95%. An aggressive target is > 90%.

Service Fill Level (24 hours)

$$\text{Service Fill Level} = \frac{\text{Parts Orders Closed in 1st 24 Hours}}{\text{Total Parts Orders}} \times 100 \quad (\%)$$

(After 24 Hours)

Service Fill Level after 24 hours is a parts management efficiency indicator that quantifies the percentage of individual parts requests entered against the on-site parts warehouse for repairs (including Backlog parts requests) and filled / closed in the first 24 hours after the first call. A reflection of the level of satisfaction of on-site parts warehouse performance. We do not have sufficient data to define a Benchmark but an aggressive target is 100%.

Unavailability Parts

$$\text{Unavailability PD} = (1 - \text{Availability}) \times \frac{\text{Parts Delay Downtime}}{\text{Total Downtime}} \times 100 \quad (\%)$$

Unavailability due to parts delays quantifies the impact of parts delay events on availability. There is no Benchmark associated with this metric but a reasonable target is < .5%. If unavailability (downtime) due to parts delays is significantly higher than .5%, it may signify potential problems with inventory quality / quality and/or a higher than normal percentage of unplanned downtime, i.e. the inability of the maintenance organization to detect problems in advance of failure and plan & schedule the work and associated resources accordingly. If parts inventory quality / quality is found to be an issue, it may be due either to the fact that the

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maintenance organization is not doing a good job of defining the parts inventory support requirements to the Parts Department or that the parts Department is not delivering on its obligation to support the site with the required parts.

Emergency Response Time

Emergency Response Time quantifies the average response time (in days) to satisfy parts requests that cannot be filled instantaneously. This parameter works and should be analyzed in conjunction with Instantaneous Service Fill Level. There is no Benchmark or target for this metric.

Parts Inventory Rotation

Parts Inventory Rotation is defined as the annual turnover of parts held in the on-site parts warehouse. No Benchmark is available for this parameter. Defining a realistic target for this metric is highly dependent upon site logistics of the specific operation including transportation, the capacity and design of the parts warehouse, the remoteness of the site, costs associated with carrying the inventory and the specific requirements of the site in terms of any availability guarantees that may be in place.

Emergency Orders

Emergency Orders quantifies the percentage of parts orders that are placed against the system on an emergency basis, i.e. "panic mode". The percentage of Emergency Orders is another method of analyzing the extent to which the maintenance organization is behaving pro-actively and control of the fleet. There is no Benchmark or target for this metric.

Inventory (items)

This metric quantifies the number of individual line items maintained on-site in the parts inventory. Because this is proportional to the size of the fleet being supported, there is no Benchmark or target for this metric. Trending inventory levels over a 6 to 12 month period and relating the trend to fleet performance results such as MTBS, MTTR and % of scheduled work, is one way of determining the impact of parts support on the overall site performance.

Inventory (value)

This metric quantifies the value of the on-site parts inventory. Here again, because this is proportional to the size of the fleet being supported, there is no Benchmark or target for this metric. Trending inventory value over a 6 to 12 month period and relating the trend to fleet performance results such as MTBS, MTTR and % of scheduled work, is one way of determining the impact of parts support on the overall site performance.

% of Backlogs waiting for Parts.

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This metric measures how many of the Backlogs are waiting on parts. This metric is found by dividing the total number of backlog orders by the number of Backlog orders that are waiting on parts. There is no benchmark

3.0 Implementation Steps

- Review Parts Management strategy established from PM process map
- Request from Regional office that they seek help from M&PSD Facility Planning group to provide design assistance
- Locate parts warehouse close to repair shop
- Employee qualified staff with Parts management background
- Review Planning/Scheduling and Backlog Management Best Practices to identify integration of processes
- Implement metrics and monitor to insure Parts Operations efficiency

4.0 Benefits

- Increased machine availability
- Efficient managed Parts Department
- Balanced parts inventory
- Minimum parts inventory cost
- Lower cost per ton and cost per hour

5.0 Resources Required

- Parts manager who understands inventory control and warehouse management
- Parts management system for storing and ordering parts
- Good communications between departments
- Familiarization of Contamination Control standards
- Parts Management process map
- Warehouse personnel
- Adequate warehouse space
- Transportation
- Storage Bins, Racks, Retrieval Equipment

6.0 Supporting Attachments / References

- Metrics (KPI's) To Assess Process Performance.pdf



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7.0 Related Best Practices

- Maintenance & Repair Processes 1007-2.0-1100
- Preventive Maintenance -1101
- Condition Monitoring -1102
- Backlog Management -1103
- Planning & Scheduling -1104
- Component Management -1106
- Repair Management -1107
- Human Resource / Training -1108
- Performance Evaluation -1109
- Continuous Improvement -1110

8.0 Acknowledgements

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